

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

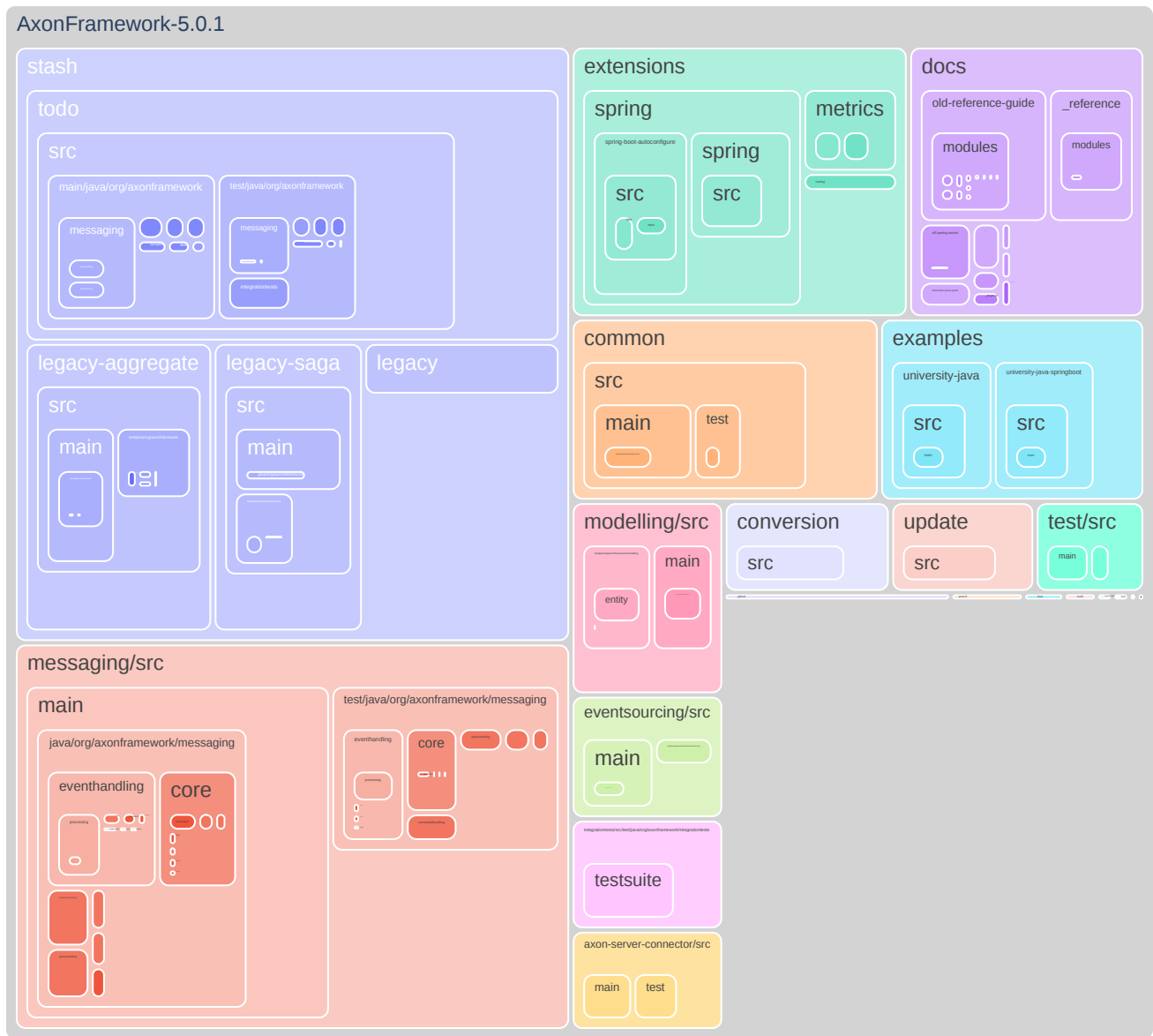
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	704.000000	704.000000	704.000000	704.000000	704.000000	704.000000
mean	22.691761	11.548295	171.423295	87.127841	144.494318	107.494318
std	112.338014	12.461968	362.848141	145.355171	200.590721	154.844381
min	1.000000	2.000000	4.000000	34.000000	35.000000	33.000000
25%	2.000000	4.000000	20.000000	34.000000	49.000000	39.000000
50%	4.000000	7.000000	40.000000	34.000000	60.000000	49.000000
75%	12.000000	14.000000	162.250000	34.000000	77.000000	60.000000
max	2594.000000	115.000000	5942.000000	524.000000	2731.000000	578.000000

	directoryPath	directoryParentPath	directoryName
0	AxonFramework-5.0.1/.idea/copyright	AxonFramework-5.0.1/.idea	copyright
1	AxonFramework-5.0.1/.idea/inspectionProfiles	AxonFramework-5.0.1/.idea	inspectionProfile
2	AxonFramework-5.0.1/axon-framework-bom	AxonFramework-5.0.1	axon-framework-bom
3	AxonFramework-5.0.1/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-5.0.1/axon-server-connector/src/main	resources/META-INF/service
4	AxonFramework-5.0.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/command	AxonFramework-5.0.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector	command
5	AxonFramework-5.0.1/axon-server-connector/src/test/resources	AxonFramework-5.0.1/axon-server-connector/src/test	resource
6	AxonFramework-5.0.1/build/coverage-report	AxonFramework-5.0.1/build	coverage-report
7	AxonFramework-5.0.1/build/parent	AxonFramework-5.0.1/build	parent
8	AxonFramework-5.0.1/common/src/main/java/org/axonframework/common/digest	AxonFramework-5.0.1/common/src/main/java/org/axonframework/common	digest
9	AxonFramework-5.0.1/common/src/main/java/org/axonframework/common/io	AxonFramework-5.0.1/common/src/main/java/org/axonframework/common	io
10	AxonFramework-5.0.1/common/src/main/resources/META-INF/services	AxonFramework-5.0.1/common/src/main	resources/META-INF/service
11	AxonFramework-5.0.1/common/src/test/java/org/axonframework/common/lifecycle	AxonFramework-5.0.1/common/src/test/java/org/axonframework/common	lifecycle
12	AxonFramework-5.0.1/common/src/test/resources/META-INF/services	AxonFramework-5.0.1/common/src/test	resources/META-INF/service
13	AxonFramework-5.0.1/conversion/src/main/resources/META-INF/services	AxonFramework-5.0.1/conversion/src/main	resources/META-INF/service
14	AxonFramework-5.0.1/docs/_playbook/localLinks	AxonFramework-5.0.1/docs/_playbook	localLink
15	AxonFramework-5.0.1/docs/_reference/modules/ROOT/attachments	AxonFramework-5.0.1/docs/_reference/modules/ROOT	attachment
16	AxonFramework-5.0.1/docs/_reference/modules/ROOT/partials	AxonFramework-5.0.1/docs/_reference/modules/ROOT	partial
17	AxonFramework-5.0.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-5.0.1/docs/_reference/modules/axon-server-connector	page
18	AxonFramework-5.0.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-5.0.1/docs/_reference/modules/axon-server-connector	partial
19	AxonFramework-5.0.1/docs/_reference/modules/configuration/pages	AxonFramework-5.0.1/docs/_reference/modules/configuration	page
20	AxonFramework-5.0.1/docs/_reference/modules/configuration/partials	AxonFramework-5.0.1/docs/_reference/modules/configuration	partial
21	AxonFramework-5.0.1/docs/_reference/modules/disruptor/pages	AxonFramework-5.0.1/docs/_reference/modules/disruptor	page
22	AxonFramework-5.0.1/docs/_reference/modules/disruptor/partials	AxonFramework-5.0.1/docs/_reference/modules/disruptor	partial
23	AxonFramework-5.0.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-5.0.1/docs/_reference/modules/eventsourcing	page
24	AxonFramework-5.0.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-5.0.1/docs/_reference/modules/eventsourcing	partial
25	AxonFramework-5.0.1/docs/_reference/modules/legacy/pages	AxonFramework-5.0.1/docs/_reference/modules/legacy	page
26	AxonFramework-5.0.1/docs/_reference/modules/legacy/partials	AxonFramework-5.0.1/docs/_reference/modules/legacy	partial
27	AxonFramework-5.0.1/docs/_reference/modules/metrics/pages	AxonFramework-5.0.1/docs/_reference/modules/metrics	page
28	AxonFramework-5.0.1/docs/_reference/modules/metrics/partials	AxonFramework-5.0.1/docs/_reference/modules/metrics	partial
29	AxonFramework-5.0.1/docs/_reference/modules/micrometer/pages	AxonFramework-5.0.1/docs/_reference/modules/micrometer	page

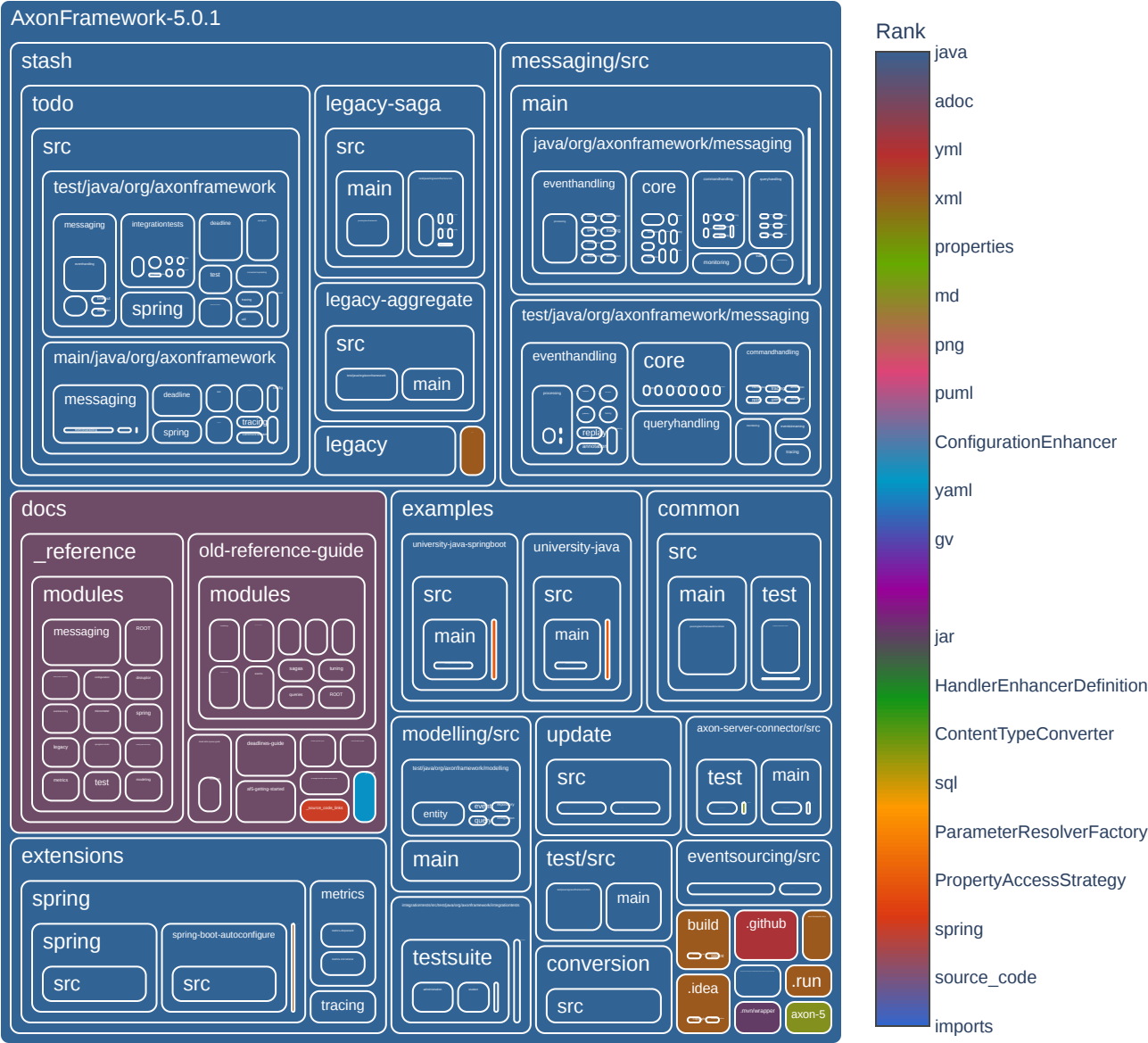
Number of files per directory

Directories and their file count



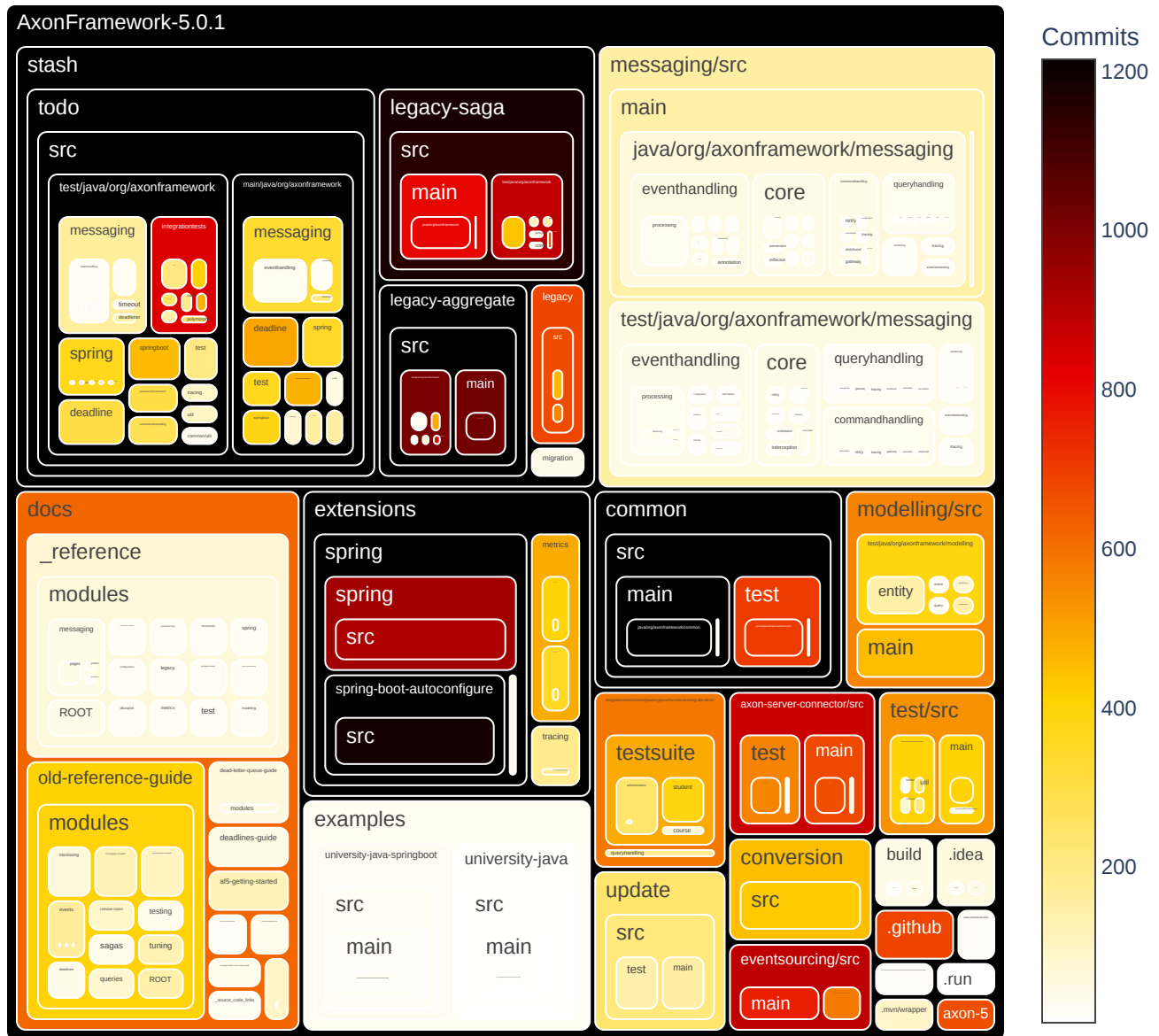
Most frequent file extension per directory

Most frequent file extension per directory



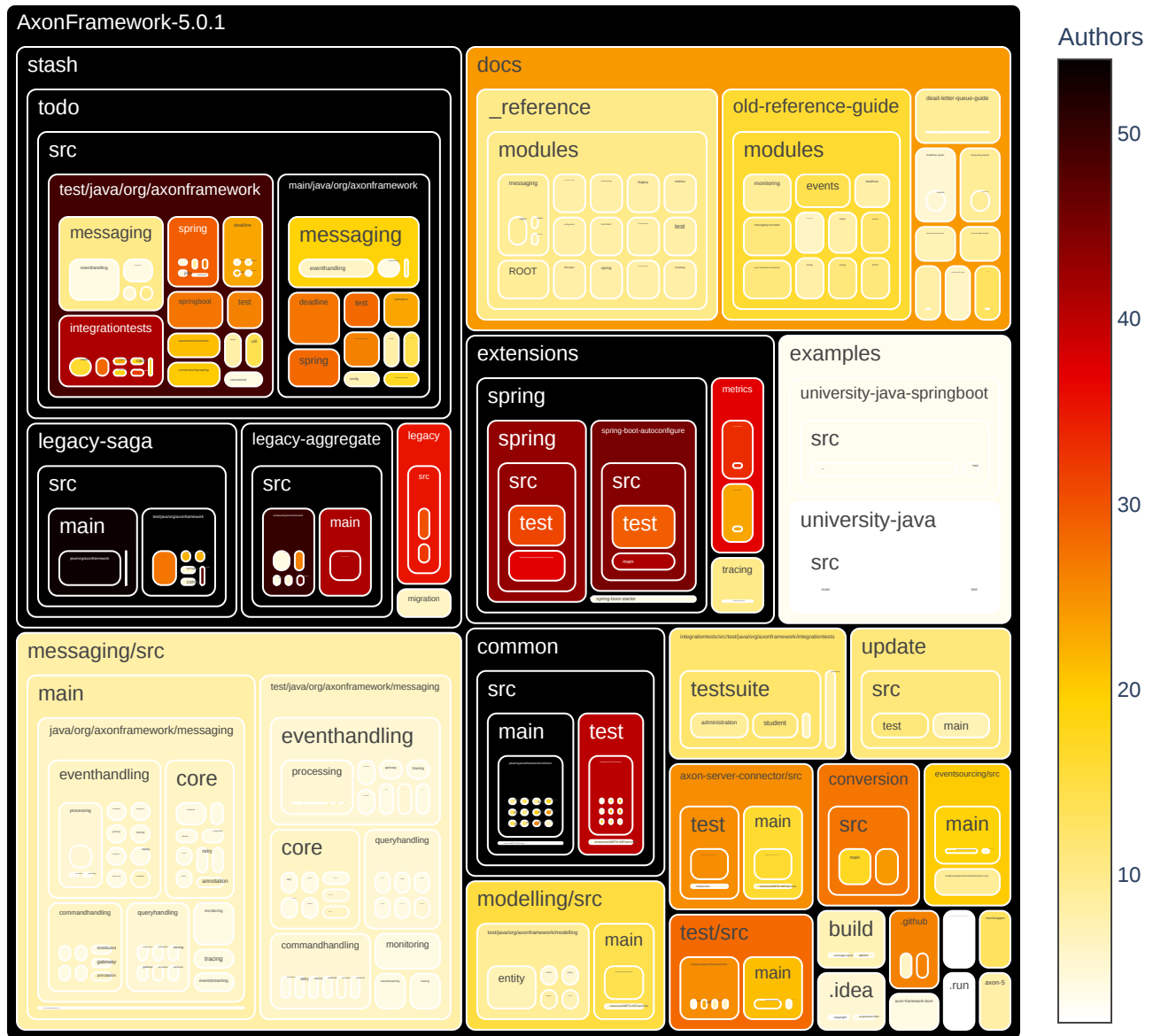
Number of commits per directory

Number of git commits



Number of distinct authors per directory

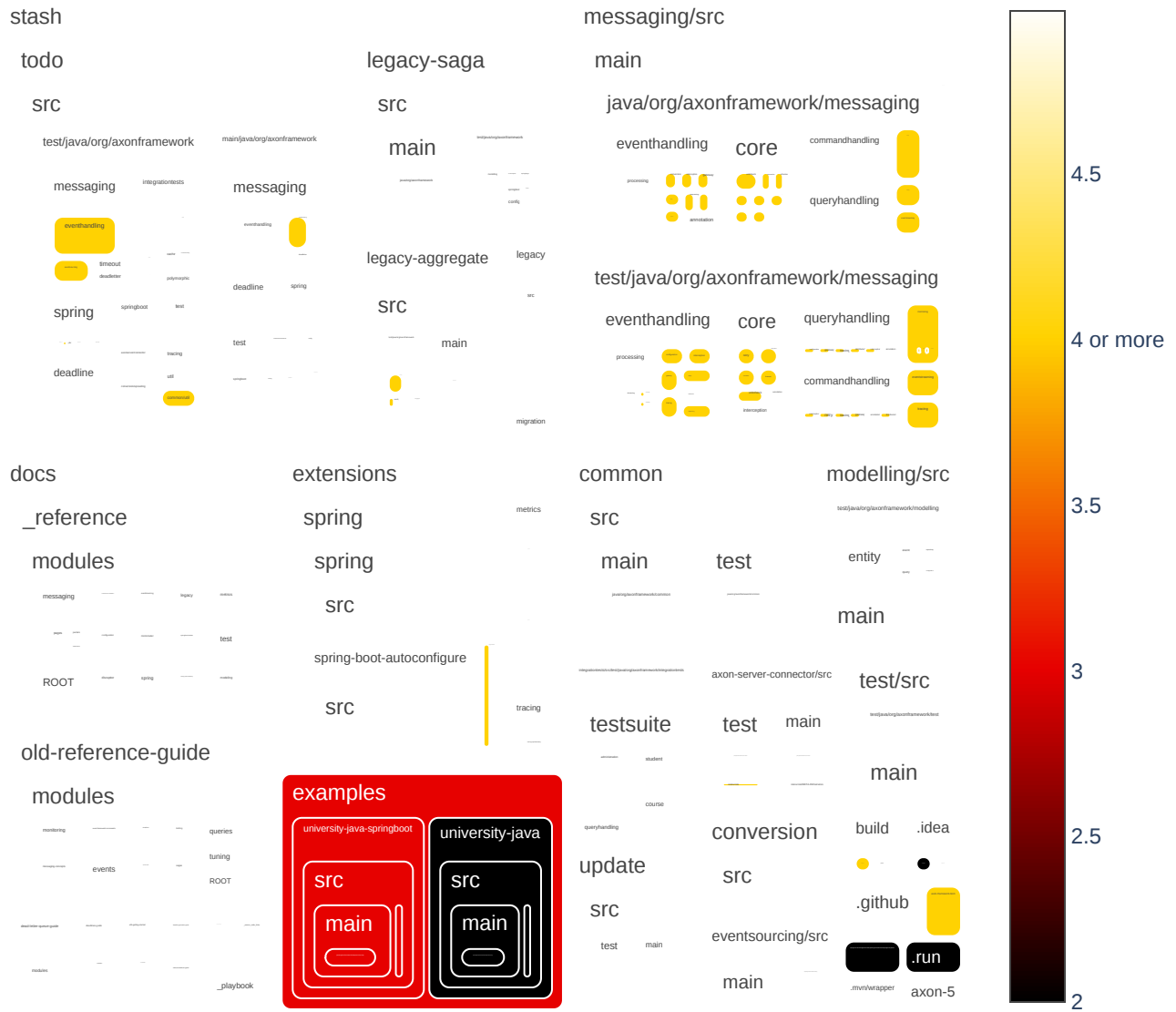
Number of distinct commit authors



Directories with very few different authors

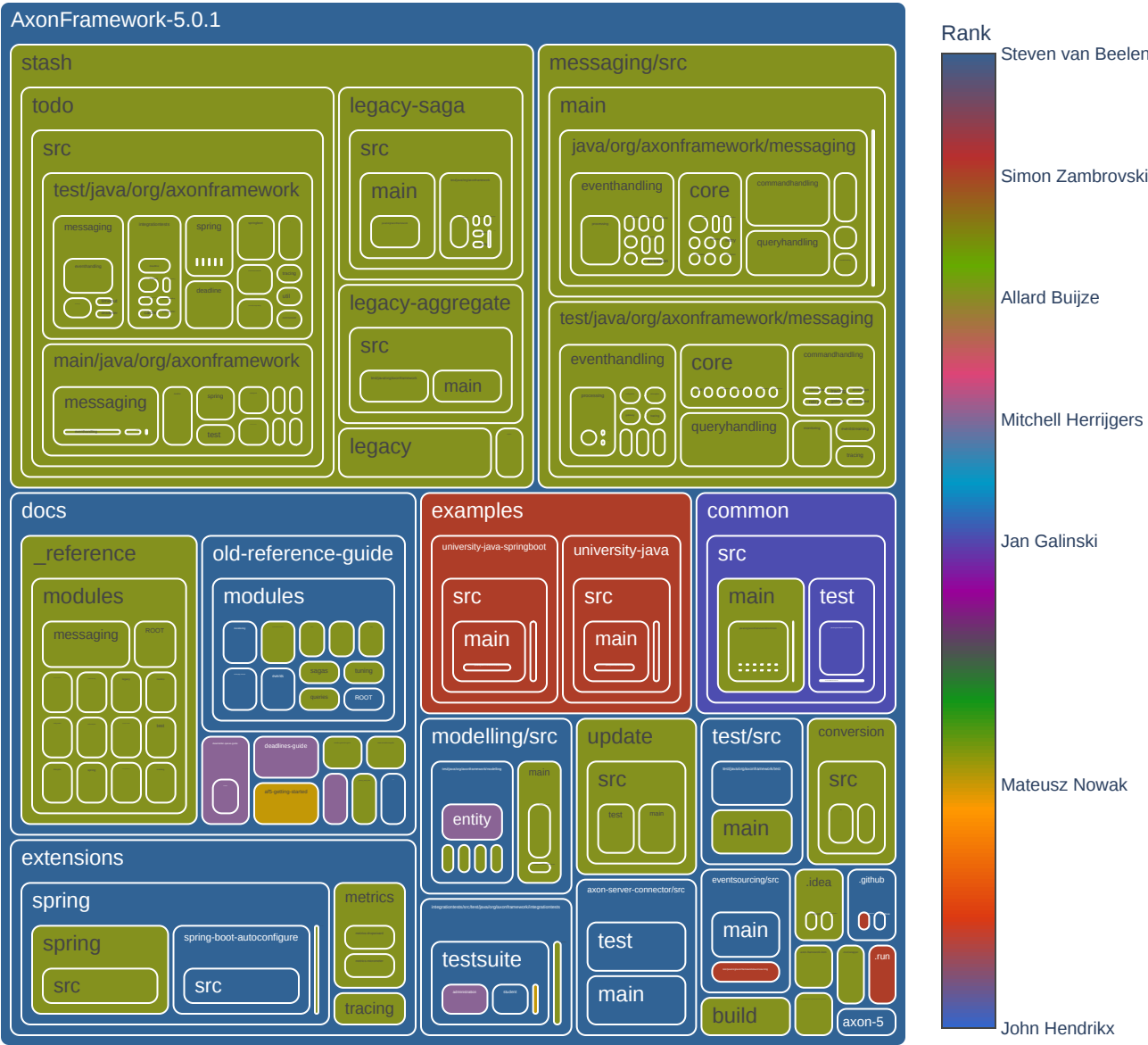
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-5.0.1



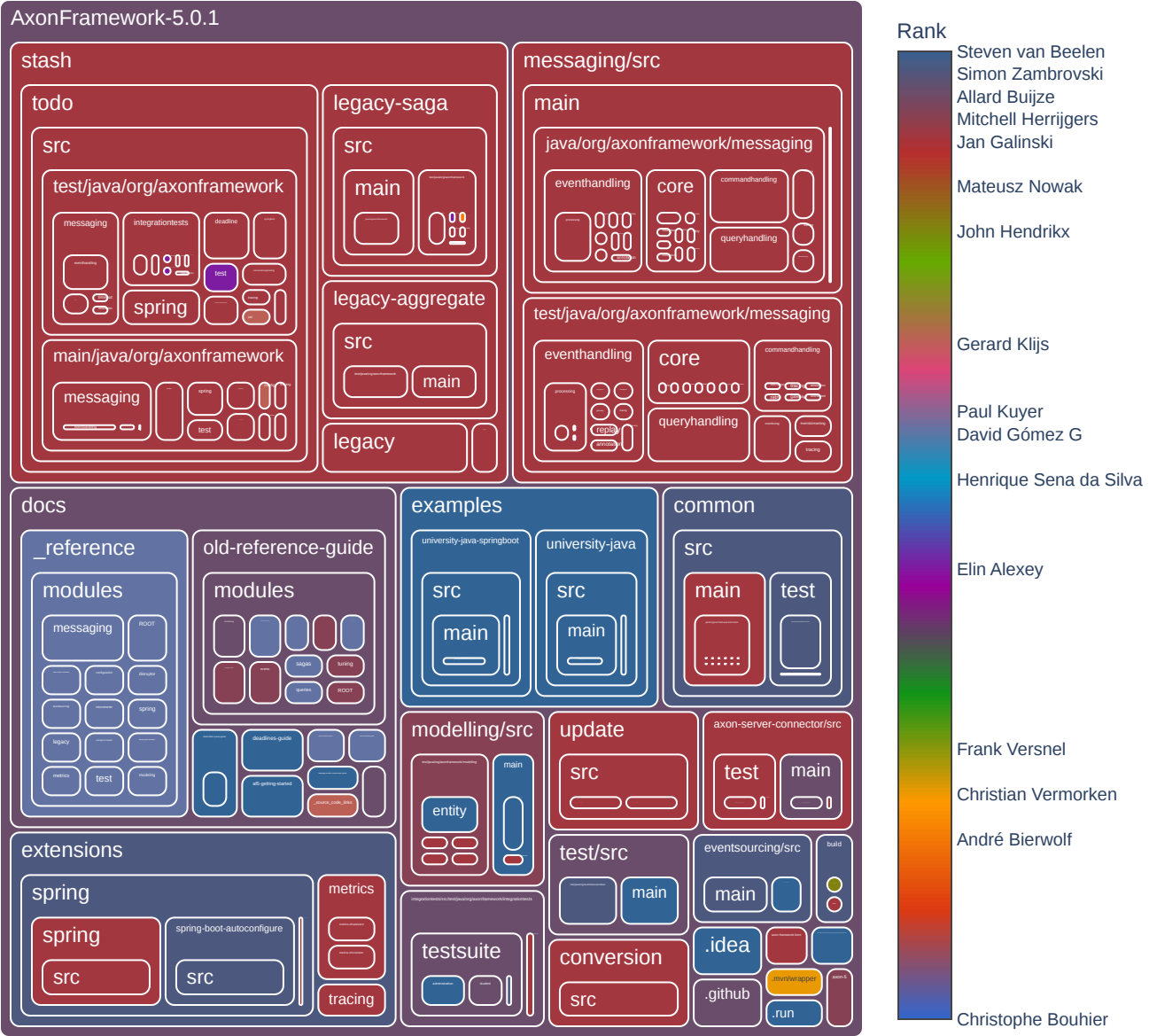
Main author per directory

Main authors with highest number of commits



Second author per directory

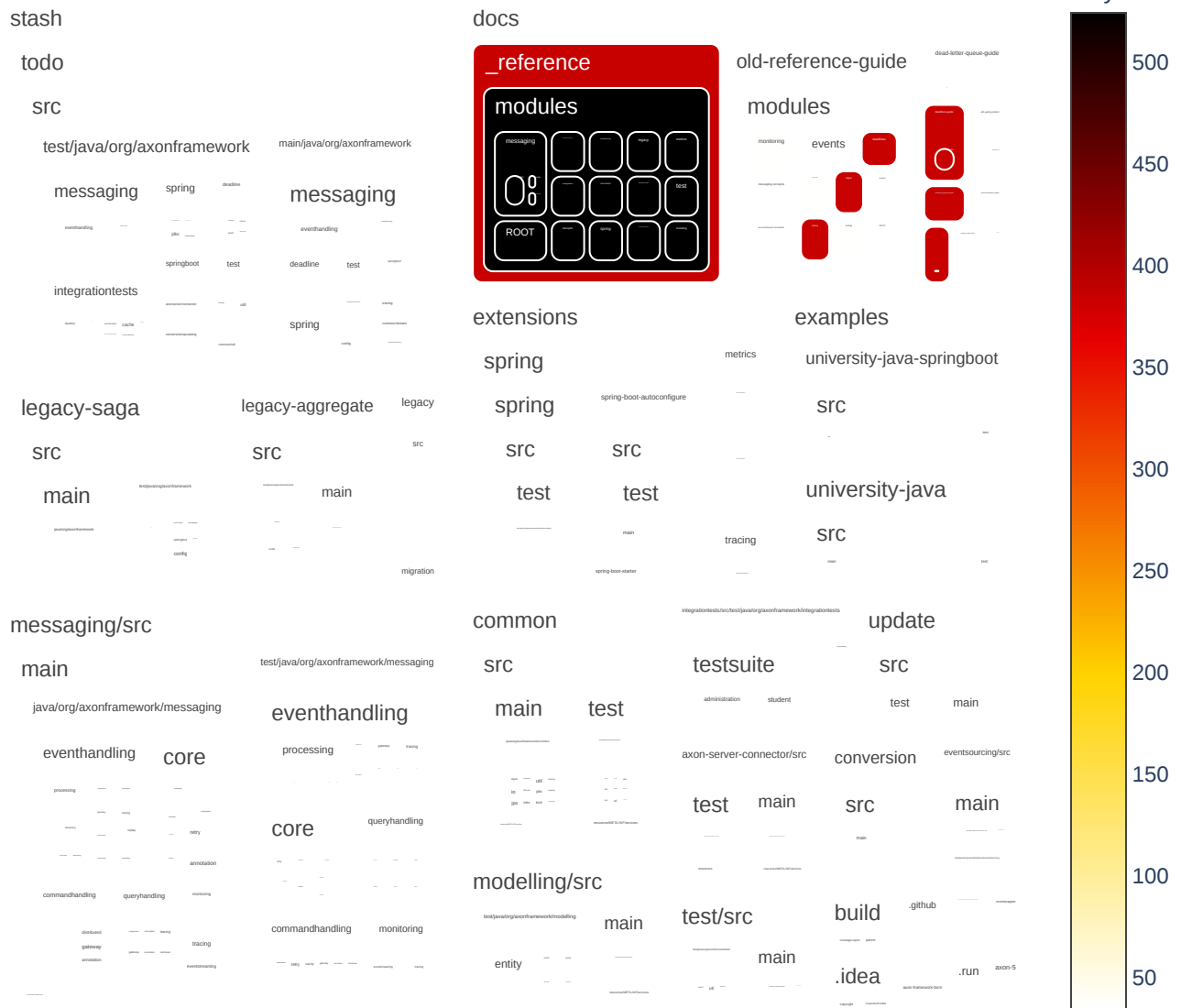
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

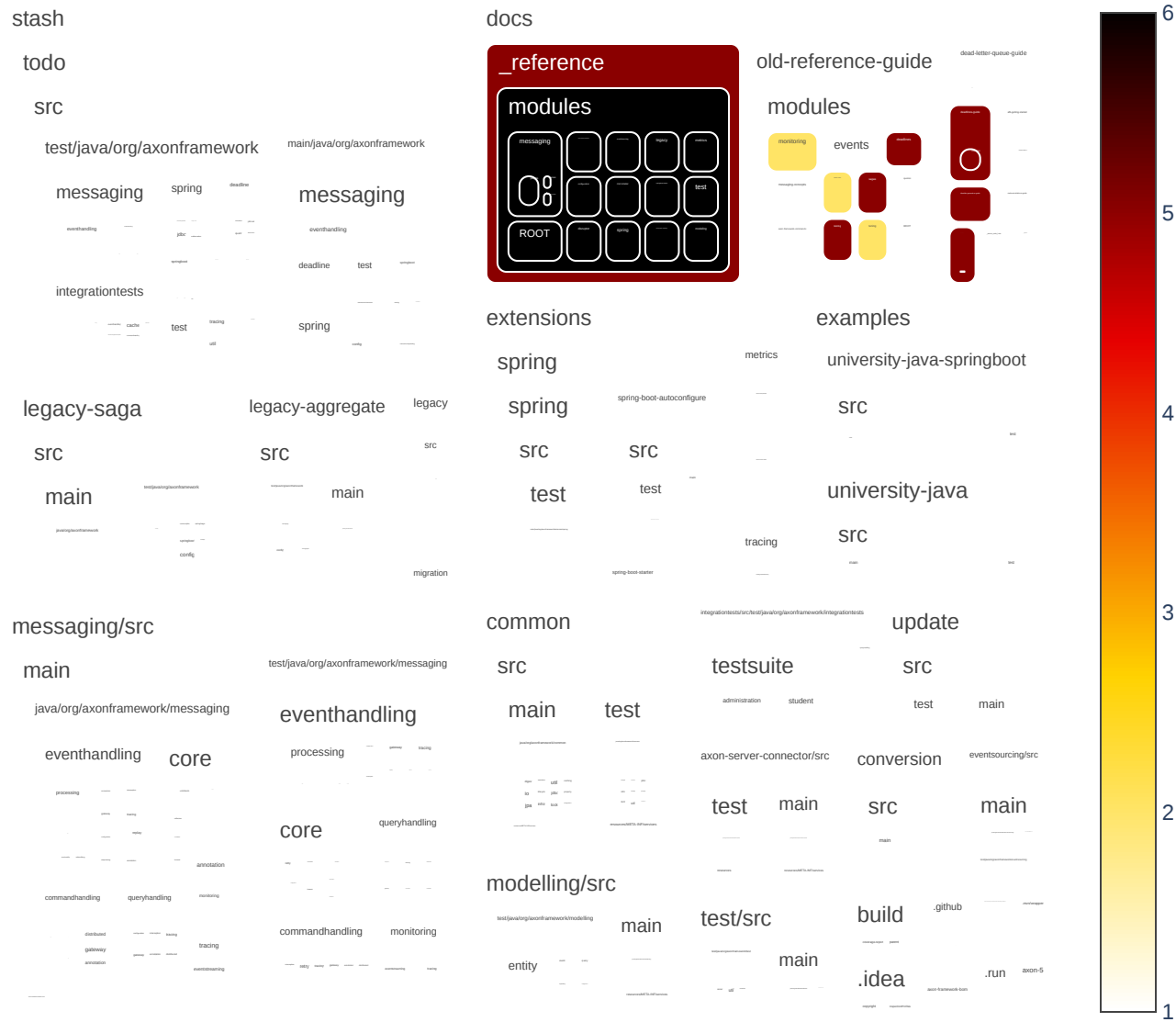
AxonFramework-5.0.1



Days since last commit per directory (ranked)

Rank of days since last commit

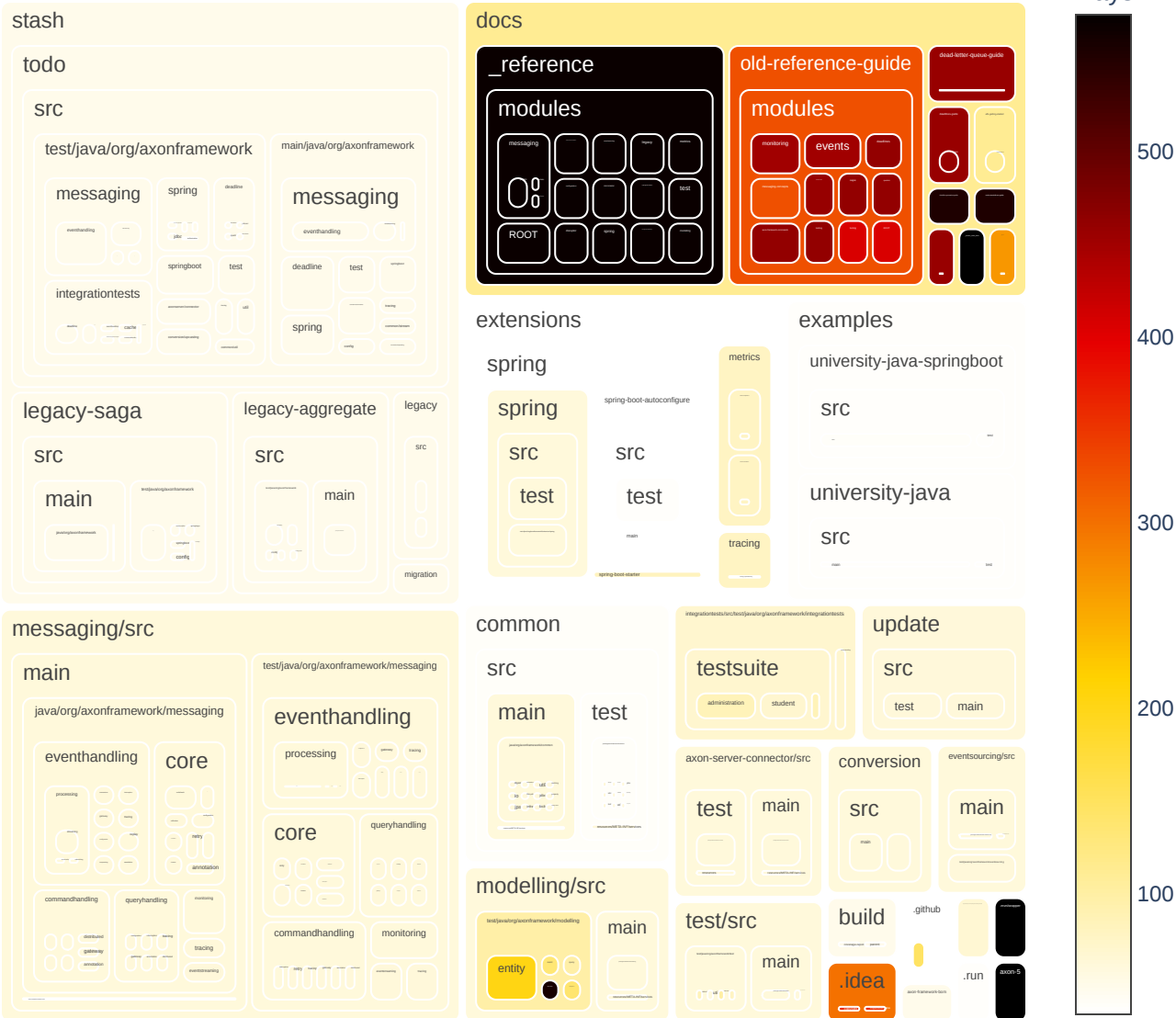
AxonFramework-5.0.1



Days since last file creation per directory

Days since last file creation

AxonFramework-5.0.1



Days since last file creation per directory (ranked)

Rank of days since last file creation

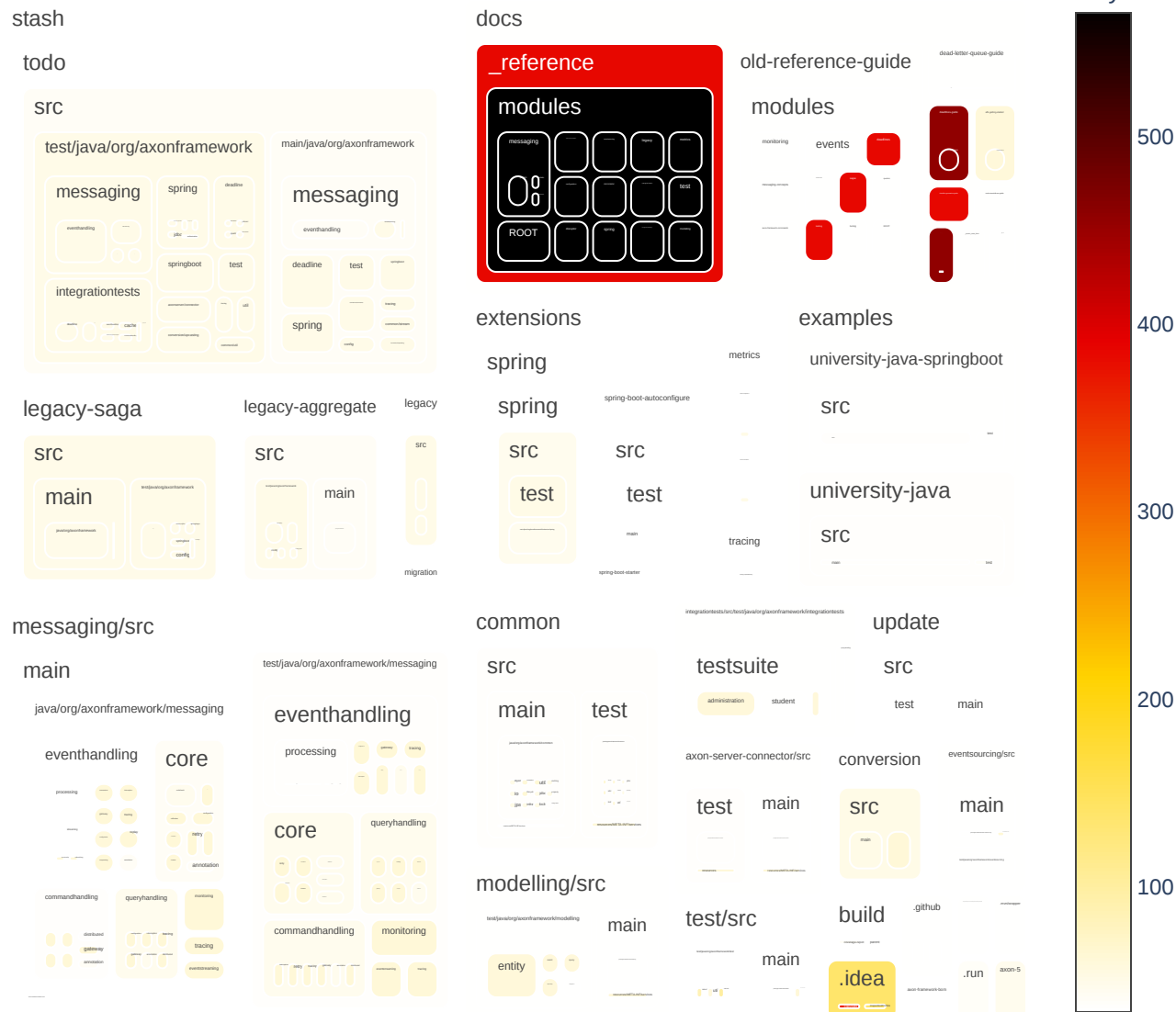
AxonFramework-5.0.1



Days since last file modification per directory

Days since last file modification

AxonFramework-5.0.1



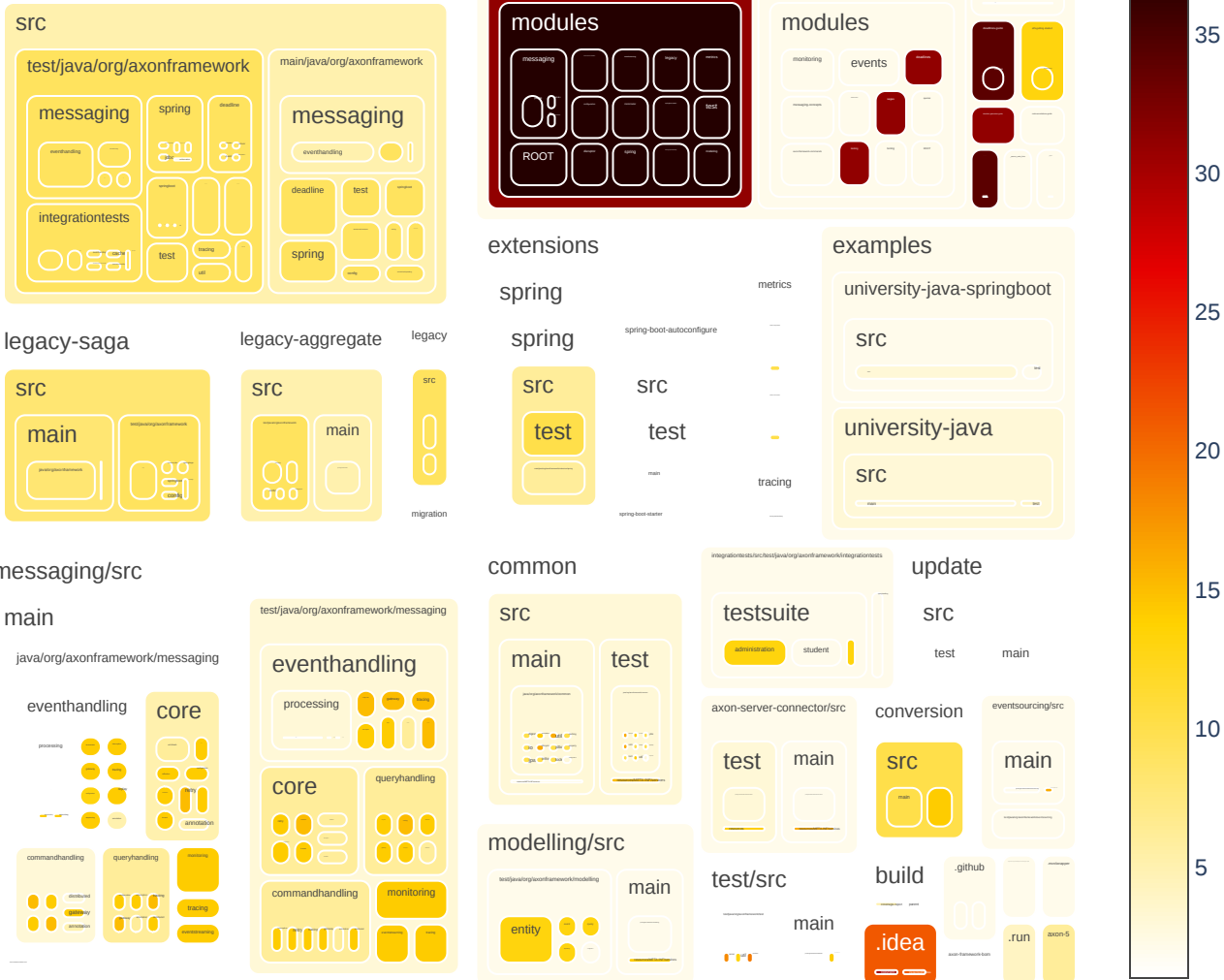
Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-5.0.1

stash

todo



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 1653

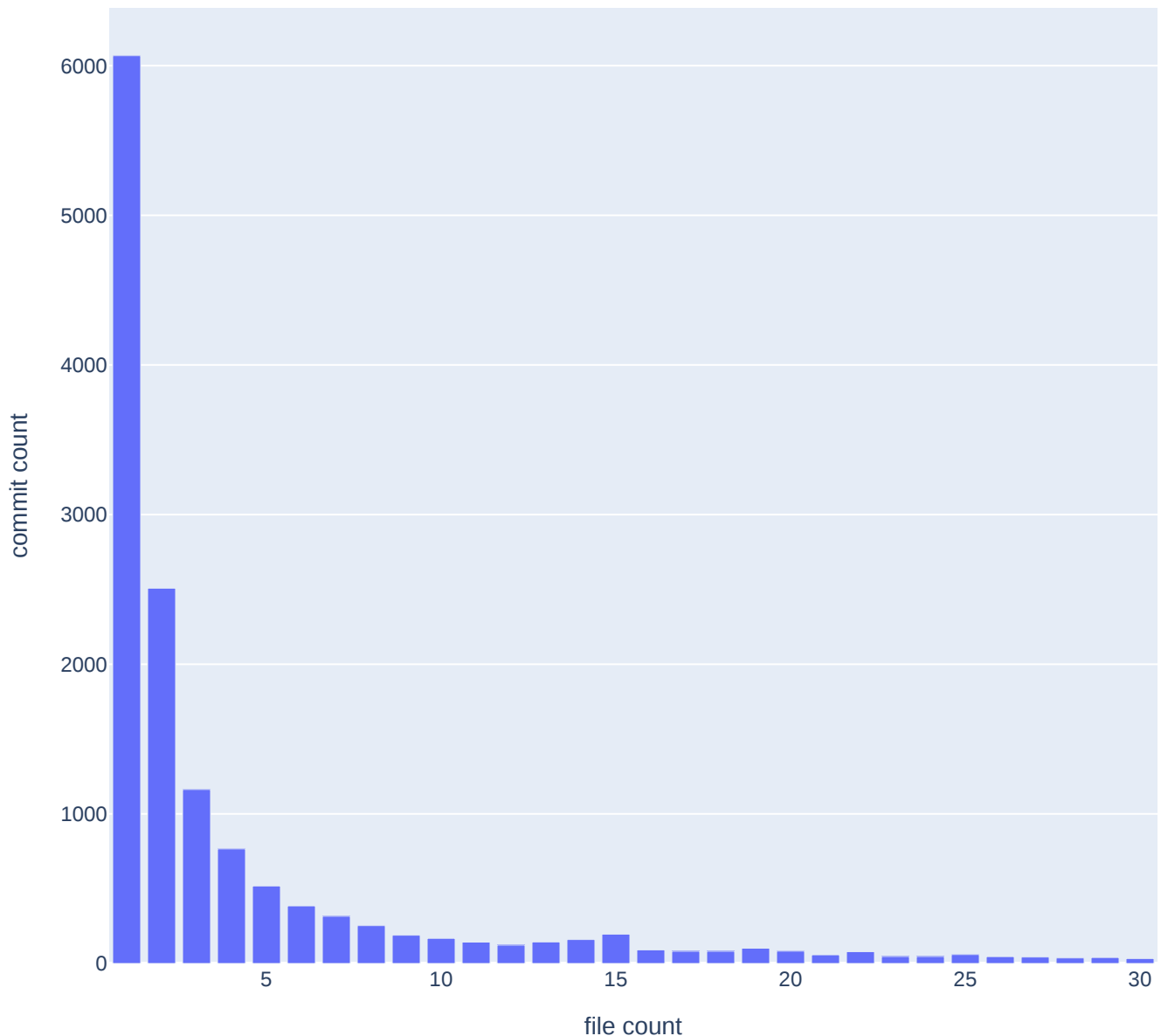
Max changed files with one commit = 4602

	filesPerCommit	commitCount
count	471.000000	471.000000
mean	546.484076	33.295117
std	759.413474	310.676325
min	1.000000	1.000000
25%	118.500000	1.000000
50%	257.000000	2.000000
75%	584.000000	5.000000
max	4602.000000	6067.000000

	filesPerCommit	commitCount
0	1	6067
1	2	2508
2	3	1163
3	4	766
4	5	518
5	6	385
6	7	316
7	8	253
8	9	189
9	10	168
10	11	143
11	12	123
12	13	144
13	14	160
14	15	196
15	16	91
16	17	82
17	18	82
18	19	102
19	20	83
20	21	58
21	22	79
22	23	47
23	24	47
24	25	60
25	26	46
26	27	43
27	28	37
28	29	40
29	30	33

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

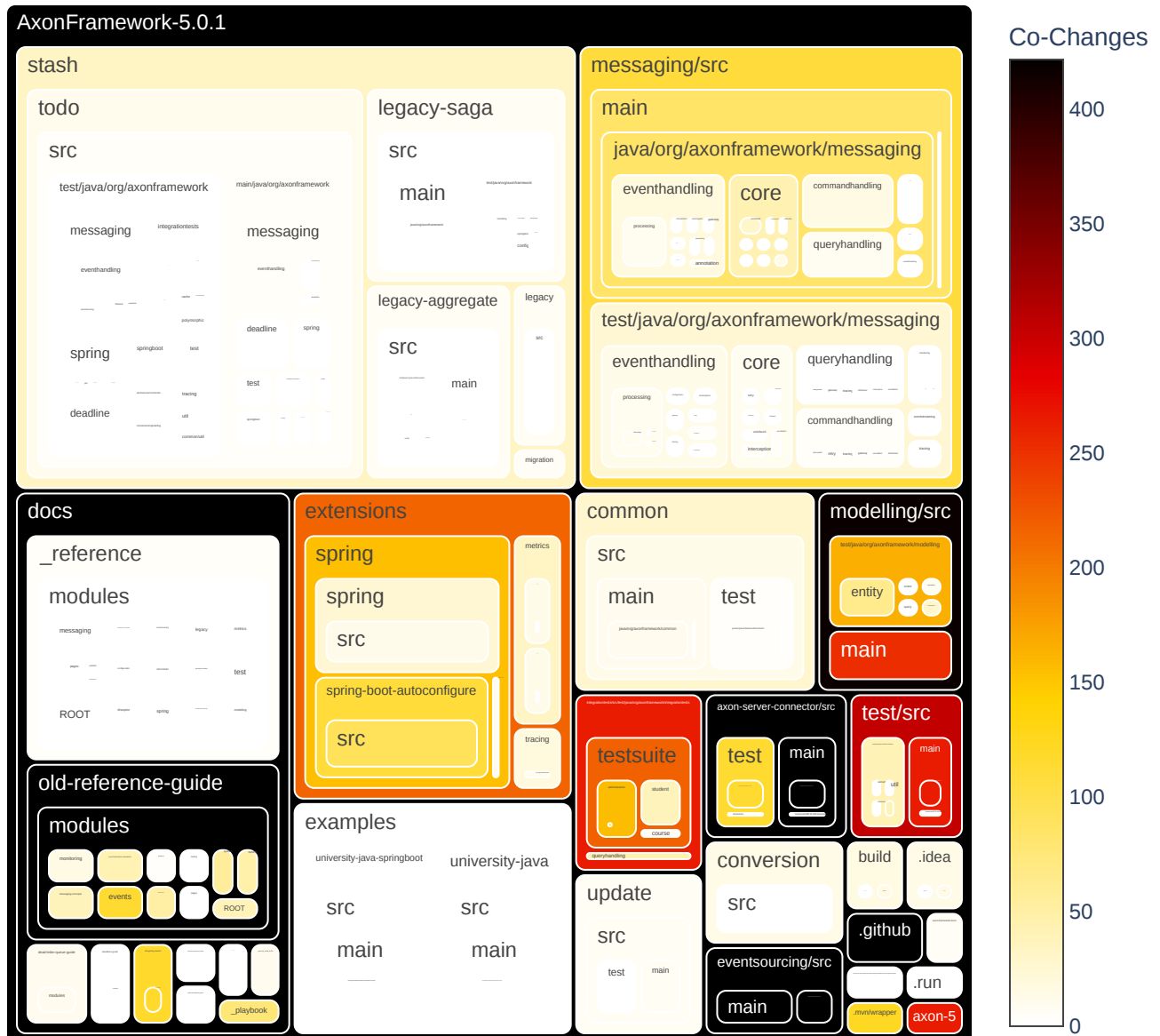
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

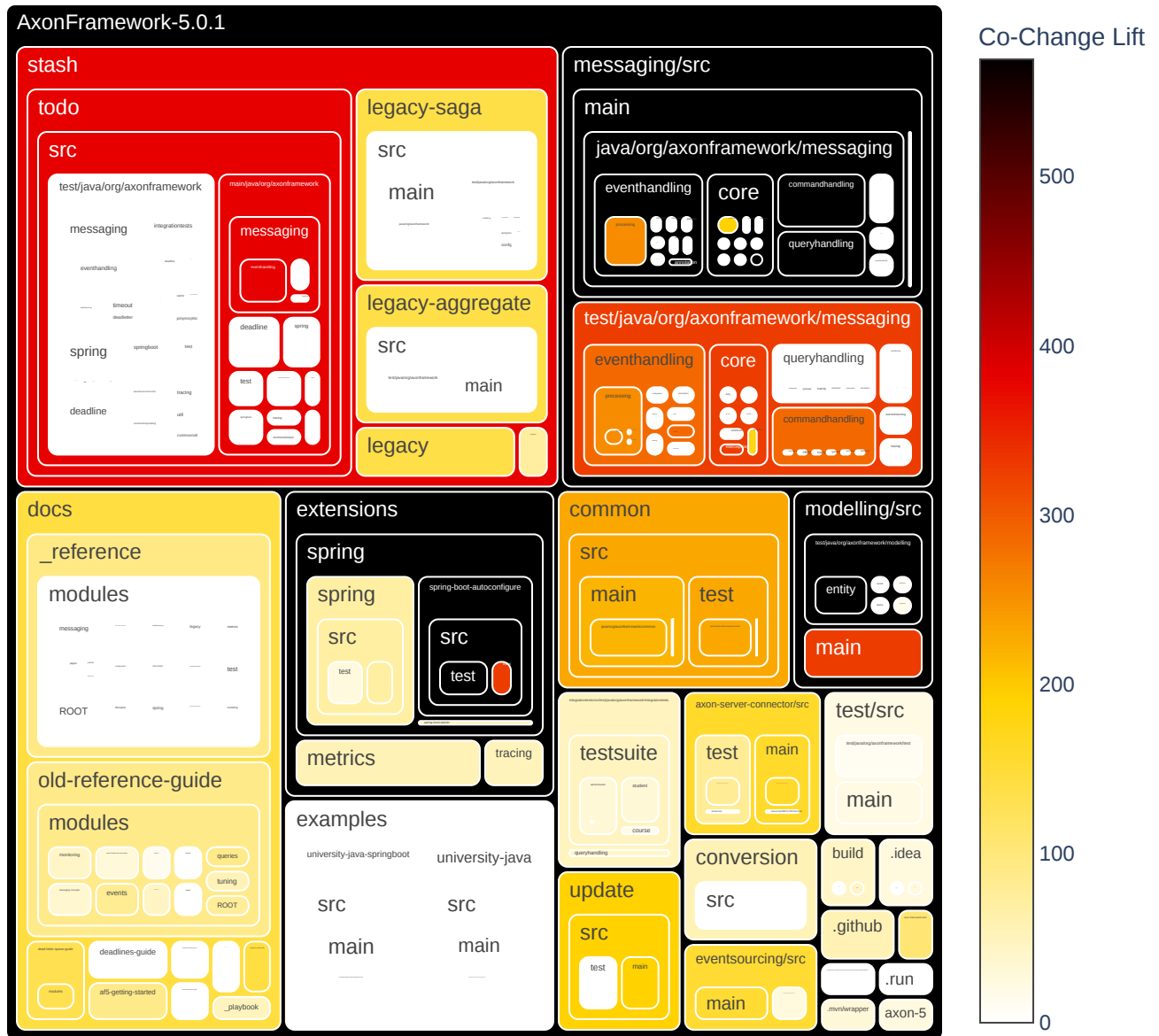
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,



	fileExtensionPair	fileExtensionPairCount
0	java↔java	2198
1	xml↔xml	190
2	adoc↔adoc	176
3	java↔md	139

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair
0	java↔java	109	1	AggregateBasedStorageEngineTestSuite↔StorageEngineTestSuite
1	java↔java	105	2	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine connector
2	java↔java	102	3	AggregateBasedJpaEventStorageEngine↔AggregateBasedStorageEngineTestSuite ev
3	java↔java	79	4	AggregateBasedAxonServerEventStorageEngine↔StorageEngineTestSuite a
4	java↔java	71	5	DefaultEventStoreTransactionTest↔StorageEngineTestSuite
5	java↔java	68	6	AggregateBasedStorageEngineTestSuite↔DefaultEventStoreTransactionTest
6	java↔java	61	7	DefaultEventStoreTransaction↔AggregateBasedStorageEngineTestSuite
7	java↔java	56	8	DefaultEventStoreTransactionTest↔SimpleEventStoreTest
8	java↔java	55	9	AggregateBasedStorageEngineTestSuite↔SimpleEventStoreTest
9	java↔java	53	10	AggregateBasedAxonServerEventStorageEngine↔DefaultEventStoreTransactionTest
10	xml↔xml	14	1	pom↔pom
11	xml↔xml	13	2	pom↔pom
12	xml↔xml	12	3	pom↔pom
13	xml↔xml	11	4	pom↔pom
14	xml↔xml	10	5	pom↔pom
15	xml↔xml	9	6	pom↔pom
16	xml↔xml	8	7	pom↔pom
17	xml↔xml	7	8	pom↔pom
18	xml↔xml	6	9	pom↔pom
19	xml↔xml	5	10	pom↔pom
20	adoc↔adoc	21	1	major-releases↔minor-releases
21	adoc↔adoc	19	2	query-dispatchers↔query-handlers
22	adoc↔adoc	18	3	streaming↔subscribing
23	adoc↔adoc	16	4	event-versioning↔event-snapshots
24	adoc↔adoc	15	5	implement-command-create-course↔implement-command-subscribe-student
25	adoc↔adoc	14	6	event-versioning↔known-issues-and-workarounds
26	adoc↔adoc	13	7	implement-command-create-course↔index
27	adoc↔adoc	12	8	streaming↔major-releases
28	adoc↔adoc	11	9	configure-axon-server↔implement-command-create-course
29	adoc↔adoc	10	10	nav↔configure-axon-server
30	java↔md	84	1	AggregateBasedStorageEngineTestSuite↔api-changes
31	java↔md	76	2	AggregateBasedJpaEventStorageEngine↔api-changes
32	java↔md	68	3	StorageEngineTestSuite↔api-changes
33	java↔md	53	4	SimpleEventStoreTest↔api-changes
34	java↔md	49	5	DefaultEventStoreTransactionTest↔api-changes
35	java↔md	46	6	QueryThreadingIntegrationTest↔api-changes
36	java↔md	39	7	DefaultEventStoreTransaction↔api-changes
37	java↔md	37	8	EventSourcingConfigurationDefaults↔api-changes
38	java↔md	34	9	AxonServerEventStorageEngine↔api-changes
39	java↔md	31	10	EventSourcingConfigurer↔api-changes

Co-Changed Files by their commit count

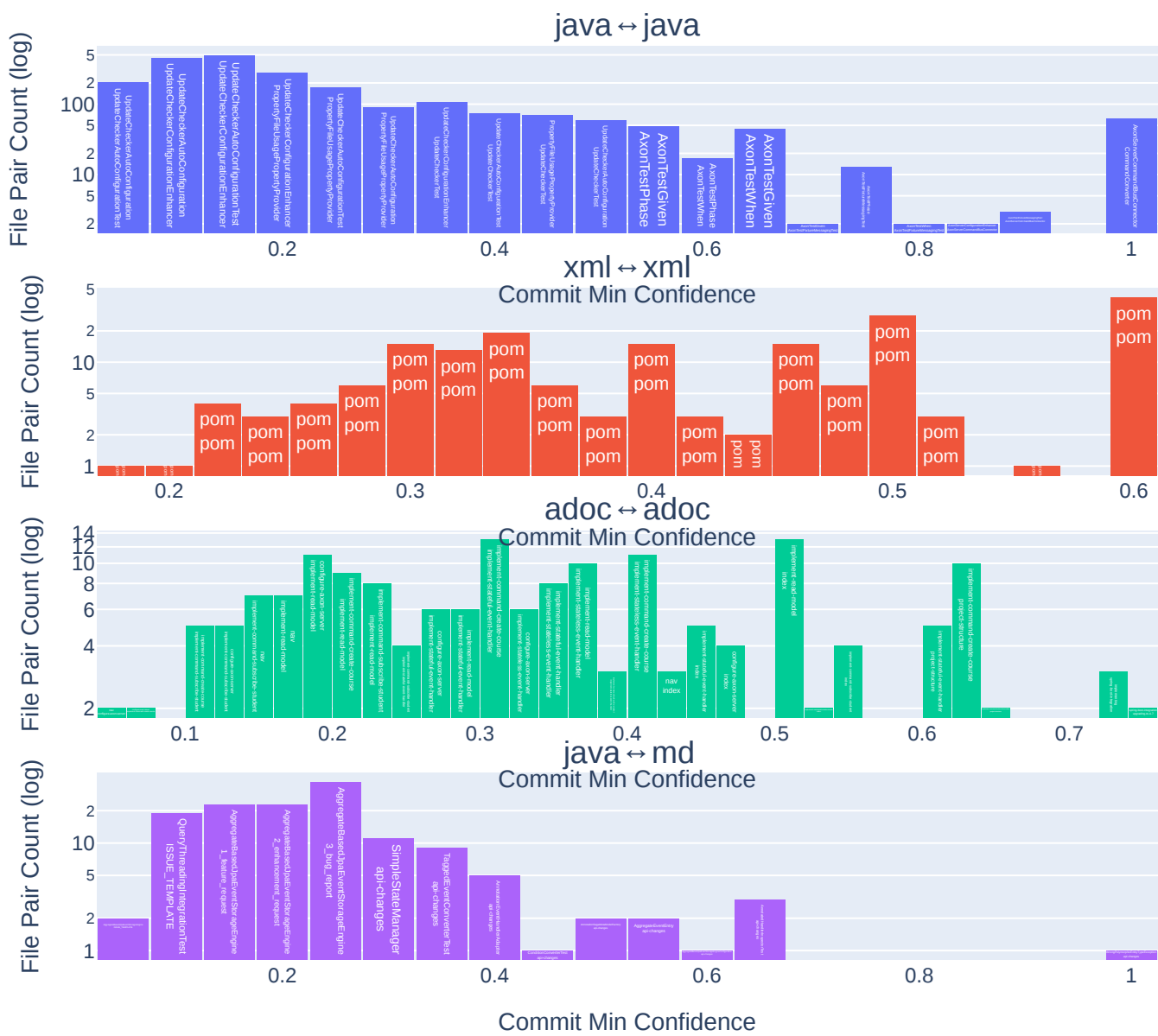


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair
0	java↔java	1.000000	1 UpdateCheckerAutoConfiguration↔UpdateCheckerAutoConfigurationTest
1	java↔java	0.900000	2 ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition
2	java↔java	0.875000	3 ChildEntityFieldDefinition↔FieldChildEntityFieldDefinition
3	java↔java	0.833333	4 AnnotatedEventHandlingComponent↔AnnotationEventHandlerAdapter messaging/src/m
4	java↔java	0.800000	5 ConditionConverter↔SpringUtils
5	java↔java	0.750000	6 UpdateCheckerConfigurationEnhancer↔UpdateCheckerTest
6	java↔java	0.714286	7 FutureUtils↔DefaultAxonApplication
7	java↔java	0.700000	8 AnnotatedEventHandlingComponent↔AnnotatedHandlerInspector me
8	java↔java	0.666667	9 DefaultAxonApplication↔DefaultAxonApplicationTest
9	java↔java	0.648148	10 DefaultAppendConditionTest↔NoAppendConditionTest
10	xml↔xml	0.600000	1 pom↔pom
11	xml↔xml	0.562500	2 pom↔pom
12	xml↔xml	0.520000	3 pom↔pom
13	xml↔xml	0.500000	4 pom↔pom
14	xml↔xml	0.480000	5 pom↔pom
15	xml↔xml	0.454545	6 pom↔pom
16	xml↔xml	0.440000	7 pom↔pom
17	xml↔xml	0.437500	8 pom↔pom
18	xml↔xml	0.423077	9 pom↔pom
19	xml↔xml	0.416667	10 pom↔pom
20	adoc↔adoc	0.750000	1 major-releases↔index
21	adoc↔adoc	0.727273	2 streaming↔dead-letter-queue
22	adoc↔adoc	0.722222	3 index↔project-structure
23	adoc↔adoc	0.650000	4 implement-command-create-course↔index
24	adoc↔adoc	0.636364	5 index↔dead-letter-queue
25	adoc↔adoc	0.625000	6 streaming↔retrying
26	adoc↔adoc	0.615385	7 configuration↔implementations
27	adoc↔adoc	0.611111	8 implement-command-create-course↔project-structure
28	adoc↔adoc	0.600000	9 spring-boot-integration↔upgrading-to-4-7
29	adoc↔adoc	0.552632	10 major-releases↔minor-releases
30	java↔md	1.000000	1 AnnotationSagaMetaModelFactory↔api-changes
31	java↔md	0.666667	2 AnnotationEventHandlerAdapter↔api-changes
32	java↔md	0.600000	3 CommandConverter↔modularity
33	java↔md	0.571429	4 ExceptionHandlerTest↔api-changes
34	java↔md	0.500000	5 AnnotatedEventHandlingComponentTest↔api-changes
35	java↔md	0.451613	6 AggregateBasedAxonServerEventStorageEngine↔api-changes
36	java↔md	0.411765	7 ChildAmbiguityException↔api-changes
37	java↔md	0.400000	8 AnnotatedEventHandlingComponent↔api-changes
38	java↔md	0.378109	9 AggregateBasedJpaEventStorageEngine↔api-changes
39	java↔md	0.375000	10 FieldChildEntityFieldDefinition↔api-changes

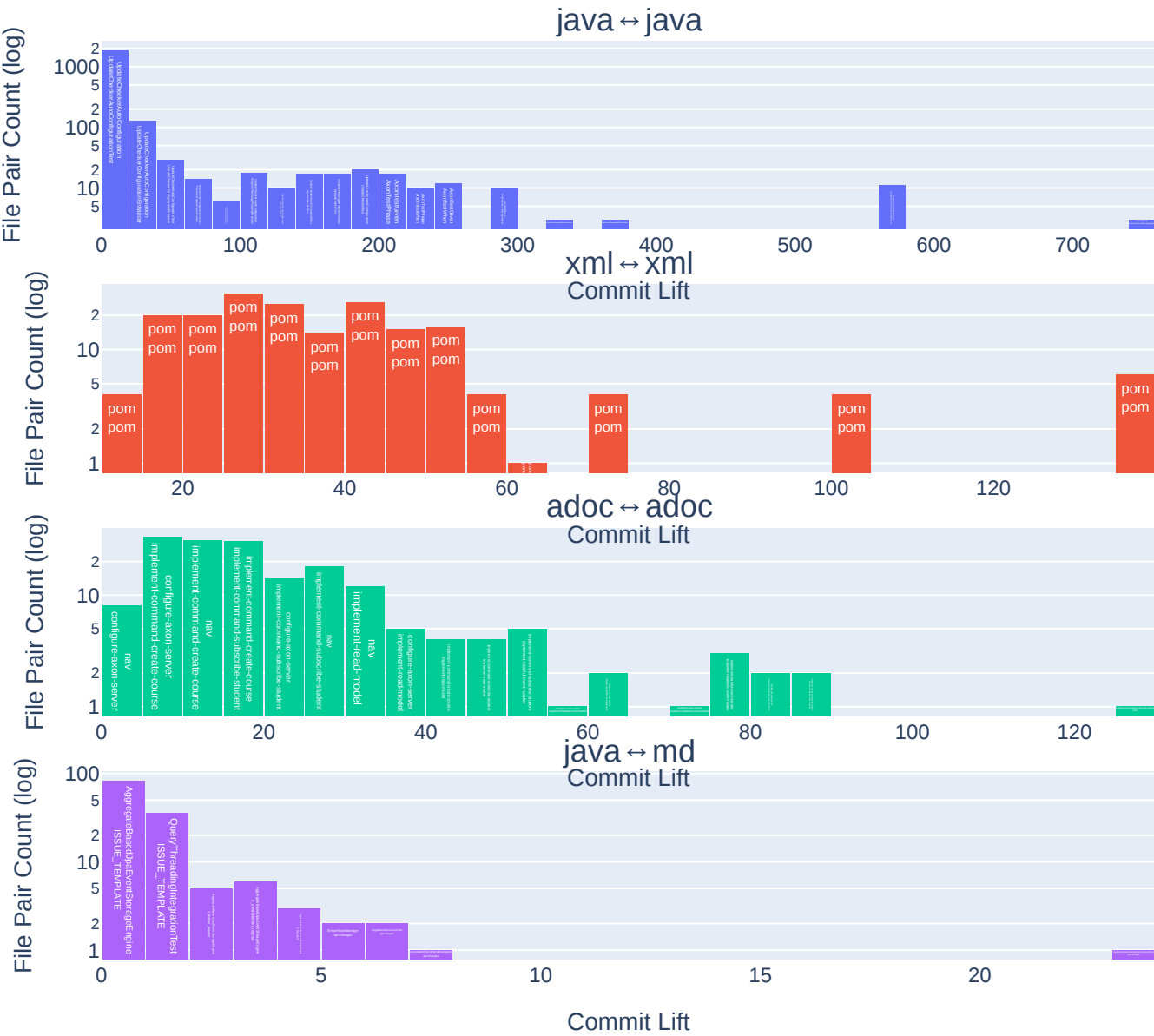
Co-Changed Files by their commit min confidence



Files changed together by commit lift

	fileExtensionPair	updateCommitLift	GroupRank	filePair	
0	java↔java	758.666667	1	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	modelling
1	java↔java	569.000000	2	UpdateCheckerAutoConfigurationTest↔UpdateCheckerTest	extensions/spring
2	java↔java	379.333333	3	HandlerComparator↔AnnotationSagaMetaModelFactory	messaging
3	java↔java	325.142857	4	AnnotationBasedEntityEvolvingComponent↔AnnotationSagaMetaModelFactory	
4	java↔java	325.142857	5	UpdateCheckerAutoConfiguration↔UpdateCheckerAutoConfigurationTest	
5	java↔java	284.500000	6	AnnotationMessageTypeResolver↔AnnotationMessageTypeResolverTest	messaging/src/n
6	java↔java	252.888889	7	MergeTask↔SplitTask	messaging/src/main/jav
7	java↔java	252.888889	8	UpdateCheckerAutoConfigurationTest↔UpdateCheckerConfigurationEnhancer	autoco
8	java↔java	243.857143	9	UpdateCheckerAutoConfiguration↔UpdateCheckerTest	extensions/spr
9	java↔java	227.600000	10	FieldChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition	mc
10	xml↔xml	136.560000	1	pom↔pom	
11	xml↔xml	103.454545	2	pom↔pom	
12	xml↔xml	71.125000	3	pom↔pom	
13	xml↔xml	64.659091	4	pom↔pom	
14	xml↔xml	56.900000	5	pom↔pom	
15	xml↔xml	54.624000	6	pom↔pom	
16	xml↔xml	49.478261	7	pom↔pom	
17	xml↔xml	47.340800	8	pom↔pom	
18	xml↔xml	45.520000	9	pom↔pom	
19	xml↔xml	44.980237	10	pom↔pom	
20	adoc↔adoc	129.318182	1	retrying↔dead-letter-queue	
21	adoc↔adoc	87.538462	2	implement-stateful-event-handler↔implement-stateless-event-handler	
22	adoc↔adoc	82.188889	3	index↔project-structure	
23	adoc↔adoc	80.464646	4	dead-letter-queue↔index	
24	adoc↔adoc	79.027778	5	retrying↔index	
25	adoc↔adoc	76.229665	6	index↔dead-letter-queue	
26	adoc↔adoc	75.866667	7	implementing↔spring-boot-integration	
27	adoc↔adoc	74.868421	8	retrying↔index	
28	adoc↔adoc	63.664336	9	configuration↔implementations	
29	adoc↔adoc	63.222222	10	index↔nav	
30	java↔md	23.544828	1	CommandConverter↔modularity	
31	java↔md	7.418514	2	NoAppendConditionTest↔design-principles	
32	java↔md	6.947497	3	NoAppendCondition↔design-principles	
33	java↔md	6.484330	4	DefaultAppendConditionTest↔design-principles	
34	java↔md	5.937391	5	QueryThreadingIntegrationTest↔README	
35	java↔md	5.835897	6	AppendCondition↔design-principles	
36	java↔md	4.955007	7	AxonServerMessageStream↔design-principles	
37	java↔md	4.168498	8	AccessSerializingRepository↔design-principles	
38	java↔md	4.040237	9	DefaultSourcingConditionTest↔design-principles	
39	java↔md	3.979021	10	SourcingConditionTest↔design-principles	

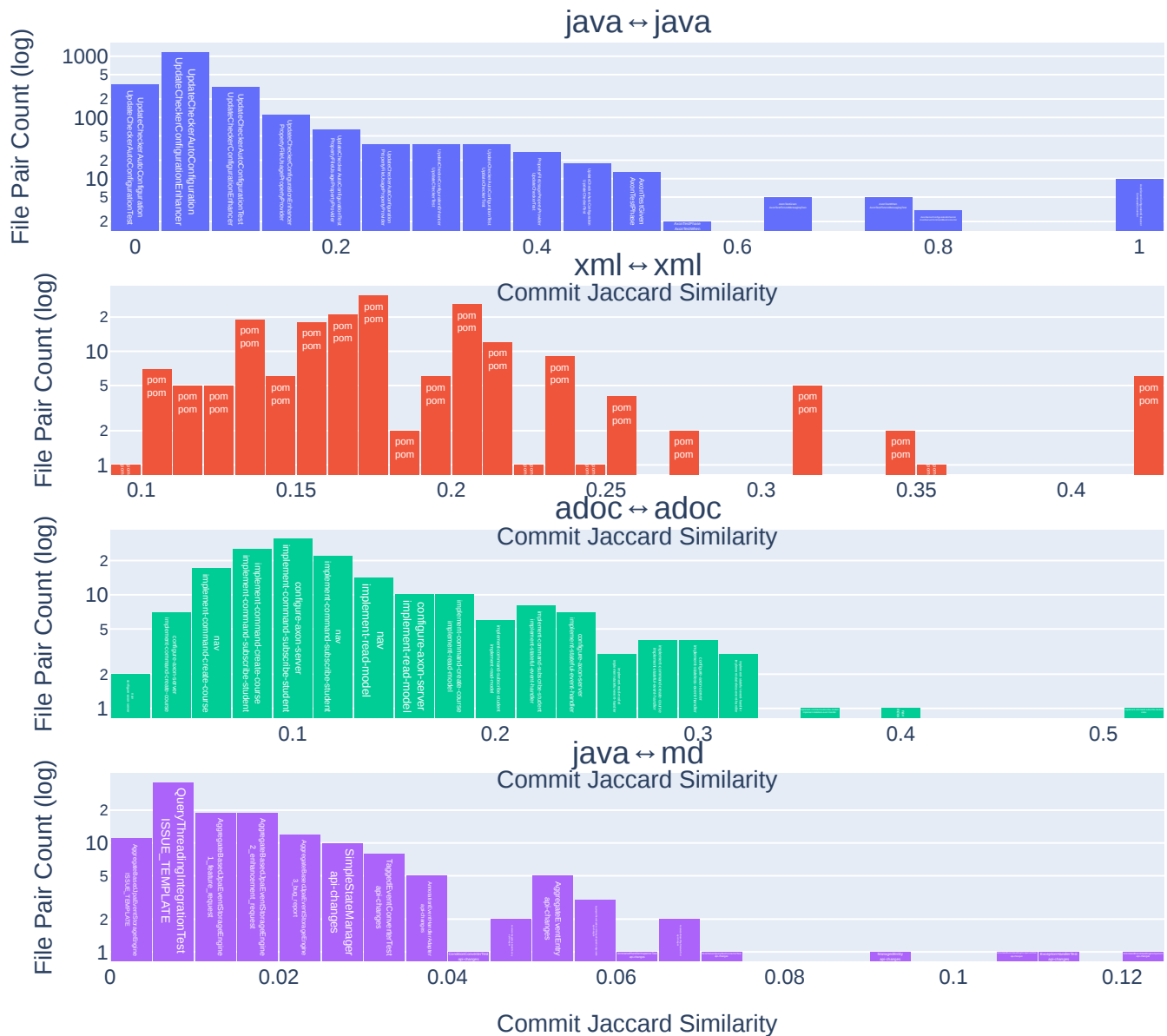
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair
0	java↔java	1.000000	1Event↔Command
1	java↔java	0.818182	2ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition
2	java↔java	0.800000	3FieldChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition
3	java↔java	0.750000	4UpdateCheckerAutoConfigurationTest↔UpdateCheckerTest
4	java↔java	0.666667	5HandlerComparator↔AnnotationSagaMetaModelFactory
5	java↔java	0.636364	6ChildEntityFieldDefinition↔FieldChildEntityFieldDefinition
6	java↔java	0.571429	7AnnotationBasedEntityEvolvingComponent↔AnnotationSagaMetaModelFactory
7	java↔java	0.500000	8MergeTask↔SplitTask
8	java↔java	0.471910	9DefaultSourcingConditionTest↔SourcingConditionTest
9	java↔java	0.461538	10AnnotatedHandlerInspector↔AnnotatedEventHandlingComponentTest
10	xml↔xml	0.428571	1pom↔pom
11	xml↔xml	0.351351	2pom↔pom
12	xml↔xml	0.342105	3pom↔pom
13	xml↔xml	0.315789	4pom↔pom
14	xml↔xml	0.312500	5pom↔pom
15	xml↔xml	0.272727	6pom↔pom
16	xml↔xml	0.250000	7pom↔pom
17	xml↔xml	0.243243	8pom↔pom
18	xml↔xml	0.238095	9pom↔pom
19	xml↔xml	0.236842	10pom↔pom
20	adoc↔adoc	0.520000	1index↔project-structure
21	adoc↔adoc	0.409091	2major-releases↔index
22	adoc↔adoc	0.357143	3retrying↔dead-letter-queue
23	adoc↔adoc	0.323529	4nav↔index
24	adoc↔adoc	0.318182	5dead-letter-queue↔index
25	adoc↔adoc	0.315789	6configuration↔index
26	adoc↔adoc	0.304348	7index↔dead-letter-queue
27	adoc↔adoc	0.300000	8index↔nav
28	adoc↔adoc	0.296875	9query-dispatchers↔query-handlers
29	adoc↔adoc	0.296296	10configuration↔implementations
30	java↔md	0.120690	1AggregateBasedAxonServerEventStorageEngine↔api-changes
31	java↔md	0.112903	2AggregateBasedStorageEngineTestSuite↔api-changes
32	java↔md	0.105702	3AggregateBasedJpaEventStorageEngine↔api-changes
33	java↔md	0.090305	4StorageEngineTestSuite↔api-changes
34	java↔md	0.074753	5SimpleEventStoreTest↔api-changes
35	java↔md	0.069014	6DefaultEventStoreTransactionTest↔api-changes
36	java↔md	0.067055	7QueryThreadingIntegrationTest↔api-changes
37	java↔md	0.062500	8NoAppendConditionTest↔design-principles
38	java↔md	0.059524	9NoAppendCondition↔design-principles
39	java↔md	0.058442	10QueryThreadingIntegrationTest↔README

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountEx
0	adoc↔adoc	index↔project-structure	14	
1	adoc↔adoc	retrying↔dead-letter-queue	25	
2	adoc↔adoc	dead-letter-queue↔index	27	
3	adoc↔adoc	major-releases↔index	27	
4	adoc↔adoc	index↔dead-letter-queue	31	
5	adoc↔adoc	configuration↔index	35	
6	adoc↔adoc	configuration↔implementations	38	
7	adoc↔adoc	nav↔index	40	
8	adoc↔adoc	retrying↔index	45	
9	adoc↔adoc	index↔nav	45	
10	java↔java	Event↔Command	54	
11	java↔java	Event↔Query	54	
12	java↔java	Command↔Query	54	
13	java↔java	Event↔QueryResponse	54	
14	java↔java	Command↔QueryResponse	54	
15	java↔java	Query↔QueryResponse	54	
16	java↔java	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition	54	
17	java↔java	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	54	
18	java↔java	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	54	
19	java↔java	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	54	
20	java↔md	AggregateBasedAxonServerEventStorageEngine↔api-changes	27	
21	java↔md	AggregateBasedJpaEventStorageEngine↔api-changes	36	
22	java↔md	AggregateBasedStorageEngineTestSuite↔api-changes	43	
23	java↔md	CommandConverter↔modularity	53	
24	java↔md	QueryThreadingIntegrationTest↔README	55	
25	java↔md	QueryThreadingIntegrationTest↔api-changes	61	
26	java↔md	SimpleEventStoreTest↔api-changes	61	
27	java↔md	AxonServerEventStorageEngine↔api-changes	68	
28	java↔md	StorageEngineTestSuite↔api-changes	68	
29	java↔md	DefaultEventStoreTransactionTest↔api-changes	74	
30	xml↔xml	pom↔pom	12	
31	xml↔xml	pom↔pom	12	
32	xml↔xml	pom↔pom	12	
33	xml↔xml	pom↔pom	12	
34	xml↔xml	pom↔pom	12	
35	xml↔xml	pom↔pom	12	
36	xml↔xml	pom↔pom	15	
37	xml↔xml	pom↔pom	17	
38	xml↔xml	pom↔pom	17	
39	xml↔xml	pom↔pom	21	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitH
0	index↔project-structure	14	13	
1	query-dispatchers↔query-handlers	79	19	
2	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine	576	105	
3	EventSourcingConfigurationDefaults↔EventSourcingConfigurationDefaultsTest	646	52	
4	NoAppendCondition↔NoAppendConditionTest	183	37	
5	DefaultAppendConditionTest↔NoAppendConditionTest	163	35	
6	DefaultSourcingConditionTest↔SourcingConditionTest	176	42	
7	AggregateBasedStorageEngineTestSuite↔StorageEngineTestSuite	739	109	
8	Event↔Command	54	4	
9	Event↔Query	54	4	
10	Command↔Query	54	4	
11	Event↔QueryResponse	54	4	
12	Command↔QueryResponse	54	4	
13	Query↔QueryResponse	54	4	
14	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition	54	8	
15	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition	64	9	
16	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	54	3	
17	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	54	3	
18	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	54	3	
19	major-releases↔index	27	18	

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCount	updateCommitH
0	java↔java	AggregateBasedJpaEventStorageEngine↔EventSourcingConfigurationDefaultsTest	3035	3	

WordCloud of git authors

	word	frequency
0	Steven van Beelen	6138
1	Allard Buijze	2917
2	smcvb	822
3	Mitchell Herrijgers	740
4	Mateusz Nowak	575
5	Milan Savic	449
6	Gerard Klijs	246
7	Marc Gathier	231
8	Simon Zambrovski	225
9	sara pellegrini	163

Wordcloud of git authors

